

Active Learning for Regression Based on Query by Committee

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Abstract. We investigate a committee-based approach for active learning of real-valued functions. This is a variance-only strategy for selection of informative training data. As such it is shown to suffer when the model class is misspecified since the learner's bias is high. Conversely, the strategy outperforms passive selection when the model class is very expressive since active minimization of the variance avoids overfitting.

1 Introduction

In process control we might wish to identify the effect of factors such as temperature, pH, etc. on output but obtaining such information, for example by running the system at various temperatures, pHs, etc., may be costly. In query learning, our goal is to provide criteria that a learning algorithm can employ to improve its performance by actively selecting data that are most informative. Given a small initial sample such a criterion might indicate that the system be run at particular temperatures, pHs, etc. in order for the relationship between these controls and the output to be better characterized.

We focus on supervised learning. Many machine learning algorithms are passive in that they receive a set of labelled data and then estimate the relationship from these data. We investigate a committee-based approach for *actively* selecting instantiations of the input variables x that should be labelled and incorporated into the training set. We restrict ourselves to the case where the training set is augmented one data point at a time, and assume that an experiment to gain the label y for an instance x is costly but computation is cheap. We investigate under what circumstances committee-based active learning requires fewer queries than passive learning.

Query by committee (QBC) was proposed by Seung, Opper and Sompolinsky [1] for active learning of classification problems. A committee of learners is trained on the available labelled data by the Gibbs algorithm. This selects a hypothesis at random from those consistent with the currently labelled data. The next query is chosen as that on which the committee members have maximal disagreement. They considered two toy models with perfectly realizable

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targets. The algorithm was implemented in the *query filtering* paradigm; the learner is given access to a stream of inputs drawn at random from the input distribution. With a two-member committee, any input on which the committee members make opposite predictions causes maximal disagreement and its label is queried. It was shown under these conditions that generalization error decreases exponentially with the number of labelled examples, but for random queries (i.e. passive learning), generalization error only decreased with an inverse power law.

Freund et al. [2] showed that QBC is an efficient query algorithm for the perceptron concept class with distributions close to uniform. This provided a rigorous proof of the earlier results, along with some relaxations in the requirements. They suggested a reasonable heuristic for filtering of queries would be to select and label those inputs expected to reduce prediction error. They note that this could be applied when the labels were not binary or even discrete. This is related to the variance-based regression methods described below. For the QBC approach to work, there must be some disagreement over the committee. In the original work, this was achieved by means of a randomized algorithm. An alternative approach is to use different subsets of the data, as in query by bagging and query by boosting [3]. This is also the approach taken in the regression framework investigated below.

2 Active Learning of Real-Valued Functions

The aim in active learning of a real-valued function is to query the labels of inputs such that the generalization error is minimized. The expected error is:

$$\int_x E_T [(\hat{y}(x; \mathcal{D}) - y(x))^2 | x] q(x) dx ,$$

where $E_T[\cdot]$ denotes expectation over $P(y|x)$ and over training sets $\mathcal{D}$, $q(x)$ is the input distribution and $\hat{y}(x; \mathcal{D})$ is the learner's output on input x , given the training set $\mathcal{D}$. The expectation in the integrand can be decomposed as [4]:

$$\begin{aligned} E_T [(\hat{y}(x; \mathcal{D}) - y(x))^2 | x] &= E [(y(x) - E[y|x])^2] + (E_{\mathcal{D}}[\hat{y}(x; \mathcal{D})] - E[y|x])^2 \\ &\quad + E_{\mathcal{D}} [(\hat{y}(x; \mathcal{D}) - E_{\mathcal{D}}[\hat{y}(x; \mathcal{D})])^2] , \end{aligned}$$

where $E_{\mathcal{D}}$ denotes expectation over training sets and the other expectations are with respect to the conditional density $P(y|x)$. The first term on the right hand side is the variance of y given x , i.e. the *noise*. The second term is the squared *bias*. The third term is the learner's *variance*. The bias and variance comprise the learner's mean squared error (MSE).

Assuming the learner is approximately unbiased, Cohn, Ghahramani and Jordan [5] propose to select inputs for labelling that minimize the variance of the learner. It is assumed that an estimate of σ_y^2 , the variance of the learner at x is available. For a new input $\tilde{x}$, an estimate of the learner's new variance at x given $\tilde{x}$ is computed. This requires an estimate of the distribution $P(\tilde{y}|\tilde{x})$, or, at least, estimates of its mean and variance. This gives an estimate of $\langle \tilde{\sigma}_{\tilde{y}}^2 \rangle$, the expected